CORE PURE / POLAR COORDINATES

Tangents to Polar Curves

Worked solutions 8 questions · 94 marks

Questions

Question 1	*6 marks*	2
Question 2	*6 marks*	4
Question 3	*17 marks*	7
Question 4	*17 marks*	11
Question 5	*10 marks*	15
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1. The curve C has polar equation

$$r = 3\sin 2\theta, \qquad 0 \leq \theta \leq \frac{\pi}{2}$$

At the non-pole point P on C, the tangent to C is perpendicular to the initial line.

Given that O is the pole, find the exact length of the line OP. **[6]**

Solution

For a polar curve,

$$x = r\cos\theta, \qquad y = r\sin\theta$$

so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}\theta}}{\frac{\mathrm{d}x}{\mathrm{d}\theta}} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$$

where $r' = \frac{\mathrm{d}r}{\mathrm{d}\theta}$.

A tangent perpendicular to the initial line is vertical, so $\frac{\mathrm{d}y}{\mathrm{d}x}$ is undefined. Therefore

$$r'\cos\theta - r\sin\theta = 0$$

with $r'\sin\theta + r\cos\theta \neq 0$

Given

$$r = 3\sin 2\theta$$

we have

$$r' = 6\cos 2\theta$$

Substituting into the vertical tangent condition gives

$$6\cos 2\theta\cos\theta - 3\sin 2\theta\sin\theta = 0$$

Now use

$$\cos 2\theta = \cos^2\theta - \sin^2\theta, \qquad \sin 2\theta = 2\sin\theta\cos\theta$$

Then

$$6(\cos^2\theta - \sin^2\theta)\cos\theta - 3(2\sin\theta\cos\theta)\sin\theta = 0$$

$$6(\cos^2\theta - \sin^2\theta)\cos\theta - 6\sin^2\theta\cos\theta = 0$$

$$6\cos\theta(\cos^2\theta - 2\sin^2\theta) = 0$$

Using $\sin^2\theta = 1 - \cos^2\theta$,

$$6\cos\theta\big(\cos^2\theta - 2(1 - \cos^2\theta)\big) = 0$$

$$6\cos\theta(3\cos^2\theta - 2) = 0$$

Since P is a non-pole point and $0 \leq \theta \leq \frac{\pi}{2}$, we must have

$$0 < \theta < \frac{\pi}{2}$$

so $\cos\theta \neq 0$

Hence

$$3\cos^2\theta - 2 = 0$$

so

$$\cos^2\theta = \frac{2}{3}$$

Also

$$\sin^2\theta = 1 - \frac{2}{3} = \frac{1}{3}$$

Because $0 < \theta < \frac{\pi}{2}$, both sine and cosine are positive, so

$$\cos\theta = \sqrt{\frac{2}{3}}, \qquad \sin\theta = \frac{1}{\sqrt{3}}$$

Now

$$\begin{aligned} r &= 3\sin 2\theta \\ &= 3(2\sin\theta\cos\theta) \\ &= 6 \cdot \frac{1}{\sqrt{3}} \cdot \sqrt{\frac{2}{3}} \\ &= 2\sqrt{2} \end{aligned}$$

Finally, check that the numerator is non-zero:

$$\begin{aligned} r'\sin\theta + r\cos\theta &= 6\cos 2\theta \sin\theta + 3\sin 2\theta\cos\theta \\ &= 6\left(\frac{1}{3}\right)\left(\frac{1}{\sqrt{3}}\right) + 3\left(\frac{2\sqrt{2}}{3}\right)\left(\sqrt{\frac{2}{3}}\right) \\ &= \frac{2}{\sqrt{3}} + \frac{4}{\sqrt{3}} \\ &= \frac{6}{\sqrt{3}} = 2\sqrt{3} \neq 0 \end{aligned}$$

So the tangent is genuinely perpendicular to the initial line, and since $OP = r$,

$$OP = 2\sqrt{2}$$

2. The curve C has polar equation

$$r = 2\sin 3\theta, \qquad 0 \leq \theta \leq \frac{\pi}{3}$$

At the non-pole point P on C, the tangent to C is parallel to the initial line.

Given that O is the pole, find the exact length of the line OP. **[6]**

Solution

For a polar curve,

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta}$$

A tangent parallel to the initial line is horizontal, so its gradient is 0. Hence we require

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$$

with $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$.

Here

$$r = 2\sin 3\theta$$

so

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = 6\cos 3\theta$$

Now

$$y = r\sin\theta$$

so, by the product rule,

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}\theta} &= \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta \\ &= 6\cos 3\theta\sin\theta + 2\sin 3\theta\cos\theta\end{aligned}$$

Set this equal to 0:

$$6\cos 3\theta\sin\theta + 2\sin 3\theta\cos\theta = 0$$

Divide by 2:

$$3\cos 3\theta\sin\theta + \sin 3\theta\cos\theta = 0$$

Using the product-to-sum identities,

$$\sin\theta\cos 3\theta = \frac{1}{2}(\sin 4\theta - \sin 2\theta), \qquad \cos\theta\sin 3\theta = \frac{1}{2}(\sin 4\theta + \sin 2\theta)$$

So

$$\begin{aligned}0 &= 3\cdot\frac{1}{2}(\sin 4\theta - \sin 2\theta) + \frac{1}{2}(\sin 4\theta + \sin 2\theta) \\ &= \frac{1}{2}\big(4\sin 4\theta - 2\sin 2\theta\big)\end{aligned}$$

Hence

$$2\sin 4\theta - \sin 2\theta = 0$$

Using $\sin 4\theta = 2\sin 2\theta \cos 2\theta$,

$$\begin{aligned} 0 &= 2(2\sin 2\theta \cos 2\theta) - \sin 2\theta \\ &= \sin 2\theta(4\cos 2\theta - 1) \end{aligned}$$

At a non-pole point on this curve, $0 < \theta < \frac{\pi}{3}$, so $0 < 2\theta < \frac{2\pi}{3}$, and therefore $\sin 2\theta \neq 0$. Thus

$$4\cos 2\theta - 1 = 0$$

so

$$\cos 2\theta = \frac{1}{4}$$

Now use $\cos 2\theta = 2\cos^2\theta - 1$:

$$\begin{aligned} 2\cos^2\theta - 1 &= \frac{1}{4} \\ 2\cos^2\theta &= \frac{5}{4} \\ \cos^2\theta &= \frac{5}{8} \end{aligned}$$

Therefore

$$\sin^2\theta = 1 - \frac{5}{8} = \frac{3}{8}$$

Since $0 < \theta < \frac{\pi}{3}$, both $\sin\theta$ and $\cos\theta$ are positive, so

$$\sin\theta = \sqrt{\frac{3}{8}} = \frac{\sqrt{6}}{4}$$

Now

$$r = 2\sin 3\theta$$

and using $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$,

$$\begin{aligned} r &= 2\left(3 \cdot \frac{\sqrt{6}}{4} - 4\left(\frac{\sqrt{6}}{4}\right)^3\right) \\ &= 2\left(\frac{3\sqrt{6}}{4} - 4 \cdot \frac{6\sqrt{6}}{64}\right) \\ &= 2\left(\frac{3\sqrt{6}}{4} - \frac{3\sqrt{6}}{8}\right) \\ &= 2\left(\frac{3\sqrt{6}}{8}\right) \\ &= \frac{3\sqrt{6}}{4} \end{aligned}$$

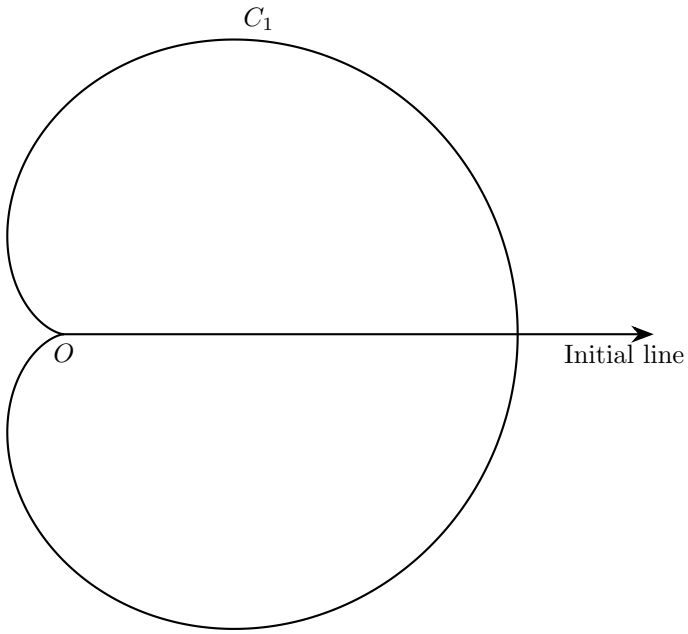
Finally, check that this really is a horizontal tangent:

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}\theta} &= \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta \\ &= 6\cos 3\theta\cos\theta - 2\sin 3\theta\sin\theta \\ &= -3 \neq 0\end{aligned}$$

So the tangent is genuinely parallel to the initial line.

Hence the exact length of OP is

$$OP = r = \frac{3\sqrt{6}}{4}$$



3. The diagram shows the sketch of a curve C_1.

The polar equation of the curve C_1 is

$$r = 2(1 + \cos\theta), \qquad -\pi \leq \theta \leq \pi$$

(a) Find the area of the region bounded by the curve C_1. **[4]**

(b) The curve C_2 whose polar equation is

$$r = \sin\theta, \qquad 0 \leq \theta \leq \pi$$

intersects the curve C_1 at the pole O and at the point A. The straight line drawn through A perpendicular to the initial line intersects the part of C_1 above the initial line again at the point B.

(i) Find the polar coordinates of A. **[4]**

(ii) Show that the length of OB is $\frac{6}{5}$. **[6]**

(iii) Find the length of AB, giving your answer to three significant figures. **[3]**

Solution

(a) Using the polar area formula,

$$\text{Area} = \frac{1}{2}\int_{-\pi}^{\pi} r^2 \, \mathrm{d}\theta$$

Here $r = 2(1 + \cos\theta)$, so

$$\text{Area} = \frac{1}{2}\int_{-\pi}^{\pi} \big(2(1 + \cos\theta)\big)^2 \, \mathrm{d}\theta = 2\int_{-\pi}^{\pi} (1 + 2\cos\theta + \cos^2\theta) \, \mathrm{d}\theta$$

Using $\cos^2\theta = \frac{1+\cos 2\theta}{2}$,

$$\text{Area} = 2\int_{-\pi}^{\pi} \left(1 + 2\cos\theta + \frac{1 + \cos 2\theta}{2}\right) \mathrm{d}\theta$$

$$= 2\int_{-\pi}^{\pi}\left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta\right)\mathrm{d}\theta$$

$$= 2\left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin 2\theta\right]_{-\pi}^{\pi}$$

At $\theta = \pi$ and $\theta = -\pi$, the sine terms are 0, so

$$\text{Area} = 2\left(\frac{3}{2}\pi - \left(-\frac{3}{2}\pi\right)\right) = 2(3\pi) = 6\pi$$

Hence the area of the region bounded by C_1 is 6π.

(b) (i) Since both curves have $r \geq 0$ on the given ranges, any intersection away from the pole must occur for the same value of θ.

So we solve

$$2(1 + \cos\theta) = \sin\theta$$

Squaring both sides,

$$4(1 + \cos\theta)^2 = \sin^2\theta$$

Using $\sin^2\theta = 1 - \cos^2\theta$,

$$4(1 + 2\cos\theta + \cos^2\theta) = 1 - \cos^2\theta$$

$$4 + 8\cos\theta + 4\cos^2\theta = 1 - \cos^2\theta$$

$$5\cos^2\theta + 8\cos\theta + 3 = 0$$

$$(5\cos\theta + 3)(\cos\theta + 1) = 0$$

So

$$\cos\theta = -\frac{3}{5} \quad \text{or} \quad \cos\theta = -1$$

The value $\cos\theta = -1$ gives $\theta = \pi$, and then $r = \sin\pi = 0$, which is the pole O.

Therefore point A corresponds to

$$\cos\theta = -\frac{3}{5}$$

Since $0 \leq \theta \leq \pi$, we have $\sin\theta > 0$, so

$$\sin\theta = \sqrt{1 - \left(-\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$$

Hence

$$r = \sin\theta = \frac{4}{5}$$

So the polar coordinates of A are

$$\left(\frac{4}{5},\ \arccos\left(-\frac{3}{5}\right)\right)$$

Equivalently, $\theta \approx 126.9°$.

(ii) The line through A perpendicular to the initial line is a vertical line, so points A and B have the same x-coordinate.

For A,

$$x_A = r\cos\theta = \frac{4}{5}\cdot\left(-\frac{3}{5}\right) = -\frac{12}{25}$$

For a general point on C_1,

$$r = 2(1+\cos\theta)$$

so

$$x = r\cos\theta = 2\cos\theta(1+\cos\theta)$$

At point B, $x = x_A$, so

$$2\cos\theta(1+\cos\theta) = -\frac{12}{25}$$

Let $c = \cos\theta$. Then

$$2c(1+c) = -\frac{12}{25}$$

$$50c^2 + 50c = -12$$

$$25c^2 + 25c + 6 = 0$$

$$(5c+3)(5c+2) = 0$$

So

$$c = -\frac{3}{5} \quad \text{or} \quad c = -\frac{2}{5}$$

The root $c = -\frac{3}{5}$ corresponds to A, so for B,

$$\cos\theta = -\frac{2}{5}$$

Hence

$$r_B = 2\left(1-\frac{2}{5}\right) = 2\cdot\frac{3}{5} = \frac{6}{5}$$

Since OB is the distance from the pole to B,

$$OB = \frac{6}{5}$$

(iii) Since AB is vertical, its length is the difference between the y-coordinates of B and A.

For A,

$$y_A = r\sin\theta = \frac{4}{5}\cdot\frac{4}{5} = \frac{16}{25}$$

For B, we have $\cos\theta = -\frac{2}{5}$, and B is above the initial line, so

$$\sin\theta = \sqrt{1-\left(-\frac{2}{5}\right)^2} = \sqrt{1-\frac{4}{25}} = \frac{\sqrt{21}}{5}$$

Therefore

$$y_B = r\sin\theta = \frac{6}{5}\cdot\frac{\sqrt{21}}{5} = \frac{6\sqrt{21}}{25}$$

So

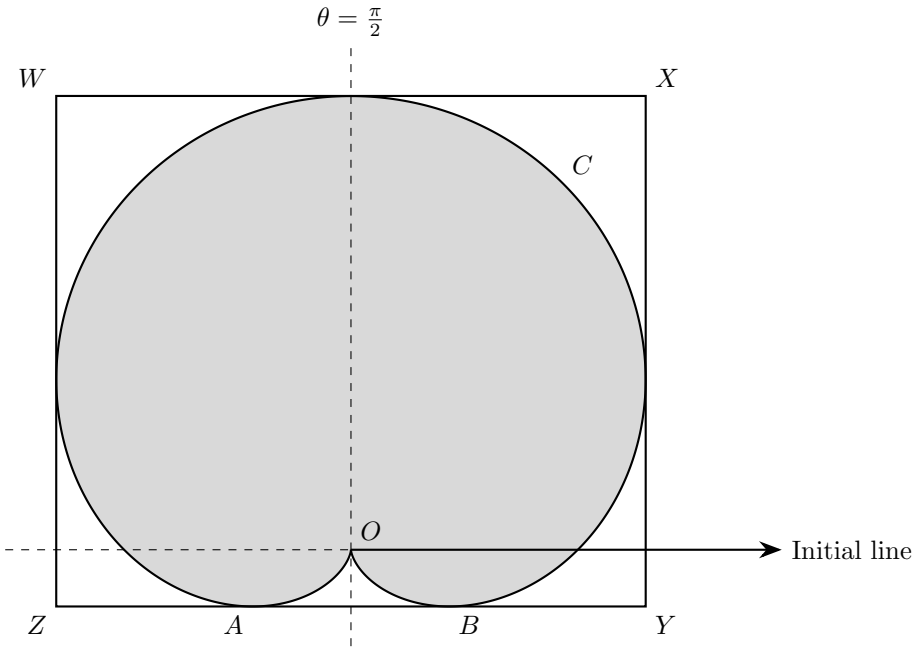
$$AB = y_B - y_A = \frac{6\sqrt{21}}{25} - \frac{16}{25} = \frac{6\sqrt{21}-16}{25}$$

$$AB \approx 0.460$$

Hence

$$AB = \frac{6\sqrt{21}-16}{25} \approx 0.460$$

to three significant figures.



4. The figure shows a sketch of the cardioid C with equation

$$r = a(1 + \sin\theta), \qquad 0 \leq \theta < 2\pi$$

Also shown are the tangents to C that are parallel and perpendicular to the initial line. These tangents form a rectangle $WXYZ$.

(a) Find the area of the finite region, shaded in the figure, enclosed by C. **[4]**

(b) Find the values of θ for which the tangent to C is parallel to the initial line. Hence find the polar coordinates of the points A and B, where ZY touches C. **[6]**

(c) Hence show that the length of WZ is $\dfrac{9a}{4}$. **[2]**

Given that the length of WX is $\dfrac{3\sqrt{3}a}{2}$

(d) find the area of the rectangle $WXYZ$. **[2]**

A logo is modelled by the cardioid C, where $a = 8\,\text{cm}$. The logo is cut from the rectangular card $WXYZ$, shown in the figure.

(e) Find a numerical value for the area of card wasted in making the logo. **[3]**

Solution

(a) For a polar curve $r = f(\theta)$, the area enclosed is

$$\frac{1}{2}\int r^2\,\mathrm{d}\theta$$

Here $r = a(1+\sin\theta)$, and $0 \le \theta < 2\pi$ traces the whole cardioid once, so

$$\begin{aligned}\text{Area} &= \frac{1}{2}\int_0^{2\pi} a^2(1+\sin\theta)^2\,\mathrm{d}\theta \\ &= \frac{a^2}{2}\int_0^{2\pi}\left(1+2\sin\theta+\sin^2\theta\right)\mathrm{d}\theta\end{aligned}$$

Using $\sin^2\theta = \frac{1-\cos 2\theta}{2}$,

$$\begin{aligned}\text{Area} &= \frac{a^2}{2}\int_0^{2\pi}\left(1+2\sin\theta+\frac{1-\cos 2\theta}{2}\right)\mathrm{d}\theta \\ &= \frac{a^2}{2}\int_0^{2\pi}\left(\frac{3}{2}+2\sin\theta-\frac{1}{2}\cos 2\theta\right)\mathrm{d}\theta \\ &= \frac{a^2}{2}\left[\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin 2\theta\right]_0^{2\pi} \\ &= \frac{a^2}{2}(3\pi)\end{aligned}$$

Therefore the area enclosed by C is

$$\frac{3\pi a^2}{2}$$

(b) Let

$$x = r\cos\theta, \qquad y = r\sin\theta$$

with

$$r = a(1+\sin\theta), \qquad \frac{\mathrm{d}r}{\mathrm{d}\theta} = a\cos\theta$$

Also,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}\theta}}{\frac{\mathrm{d}x}{\mathrm{d}\theta}}$$

A tangent parallel to the initial line is horizontal, so we need

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$$

with $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$.

Now

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}\theta} &= \frac{\mathrm{d}}{\mathrm{d}\theta}(r\sin\theta) \\ &= \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta \\ &= a\cos\theta\sin\theta + a(1+\sin\theta)\cos\theta \\ &= a\cos\theta(1+2\sin\theta)\end{aligned}$$

So

$$a\cos\theta(1+2\sin\theta) = 0$$

Hence either

$$\cos\theta = 0 \quad \text{or} \quad 1 + 2\sin\theta = 0$$

From $\cos\theta = 0$,

$$\theta = \frac{\pi}{2},\ \frac{3\pi}{2}$$

From $1 + 2\sin\theta = 0$,

$$\sin\theta = -\frac{1}{2}$$

so

$$\theta = \frac{7\pi}{6},\ \frac{11\pi}{6}$$

At $\theta = \frac{3\pi}{2}$,

$$r = a(1 + \sin\tfrac{3\pi}{2}) = a(1 - 1) = 0$$

so this is the cusp at the pole, not a smooth horizontal tangent.

Therefore the values of θ for which the tangent is parallel to the initial line are

$$\theta = \frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$$

The line ZY is the lower horizontal tangent, so it touches C at the two lower points, where

$$\theta = \frac{7\pi}{6},\ \frac{11\pi}{6}$$

At each of these,

$$r = a(1 + \sin\theta) = a\left(1 - \frac{1}{2}\right) = \frac{a}{2}$$

Hence the polar coordinates are

$$A\left(\frac{a}{2}, \frac{7\pi}{6}\right), \qquad B\left(\frac{a}{2}, \frac{11\pi}{6}\right)$$

(c) From part (b), the upper horizontal tangent occurs at $\theta = \frac{\pi}{2}$.

Then

$$r = a(1 + \sin\tfrac{\pi}{2}) = 2a$$

so its y-coordinate is

$$y = r\sin\theta = 2a \cdot 1 = 2a$$

The lower horizontal tangent touches at A and B, where $r = \frac{a}{2}$ and $\theta = \frac{7\pi}{6}$ or $\frac{11\pi}{6}$, so

$$y = \frac{a}{2}\sin\frac{7\pi}{6} = \frac{a}{2}\left(-\frac{1}{2}\right) = -\frac{a}{4}$$

Therefore the vertical distance between these tangents is

$$WZ = 2a - \left(-\frac{a}{4}\right) = \frac{9a}{4}$$

(d) The area of the rectangle is

$$\text{Area} = WX \times WZ$$

Given

$$WX = \frac{3\sqrt{3}\,a}{2} \qquad \text{and} \qquad WZ = \frac{9a}{4}$$

we get

$$\begin{aligned}\text{Area} &= \frac{3\sqrt{3}\,a}{2} \times \frac{9a}{4} \\ &= \frac{27\sqrt{3}\,a^2}{8}\end{aligned}$$

Therefore the area of rectangle $WXYZ$ is

$$\frac{27\sqrt{3}\,a^2}{8}$$

(e) When $a = 8$,

$$\begin{aligned}\text{Area of rectangle} &= \frac{27\sqrt{3}\,a^2}{8} \\ &= \frac{27\sqrt{3} \cdot 8^2}{8} \\ &= 216\sqrt{3}\ \text{cm}^2\end{aligned}$$

From part (a),

$$\begin{aligned}\text{Area of cardioid} &= \frac{3\pi a^2}{2} \\ &= \frac{3\pi \cdot 8^2}{2} \\ &= 96\pi\ \text{cm}^2\end{aligned}$$

So the wasted area is

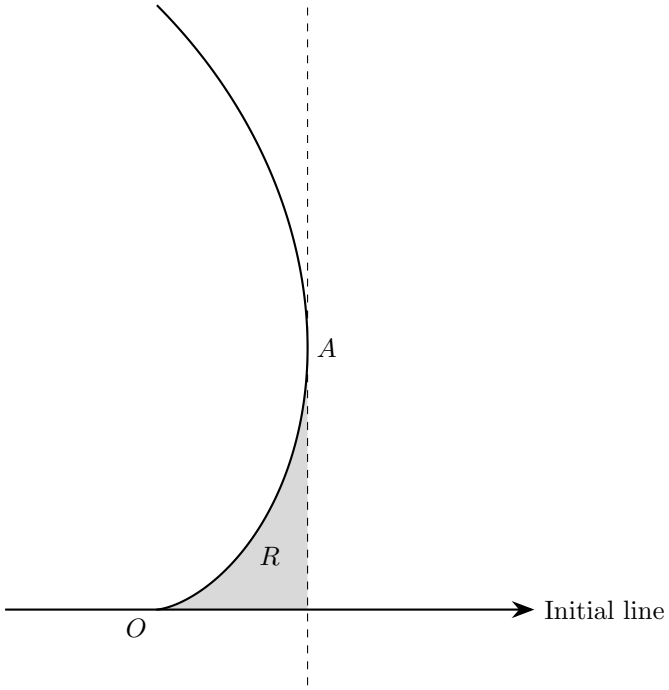
$$216\sqrt{3} - 96\pi\ \text{cm}^2$$

Numerically,

$$216\sqrt{3} - 96\pi \approx 72.5$$

Therefore the area of card wasted is

$$216\sqrt{3} - 96\pi\ \text{cm}^2 \approx 72.5\ \text{cm}^2$$



5. The diagram above shows a sketch of the curve C with equation

$$r = 4(1 - \cos\theta) \qquad 0 \leq \theta \leq \frac{\pi}{2}$$

The diagram above also shows the tangent to C at the point A, where A is not the pole.
This tangent is perpendicular to the initial line.

(a) Use differentiation to prove that the polar coordinates of A are $\left(2, \frac{\pi}{3}\right)$. **[4]**

The finite region R, shown shaded in the diagram above, is bounded by C, the tangent at A and the initial line.

(b) Use calculus to show that the exact area of R is $\frac{1}{2}(15\sqrt{3} - 8\pi)$. **[6]**

Solution

(a) For a polar curve,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\dfrac{\mathrm{d}y}{\mathrm{d}\theta}}{\dfrac{\mathrm{d}x}{\mathrm{d}\theta}}$$

Here

$$x = r\cos\theta = 4(1 - \cos\theta)\cos\theta, \qquad y = r\sin\theta = 4(1 - \cos\theta)\sin\theta$$

The initial line is the positive x-axis, so a tangent perpendicular to the initial line is a vertical tangent.
Therefore we need

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0 \quad \text{and} \quad \frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$$

Now

$$x = 4(\cos\theta - \cos^2\theta)$$

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 4(-\sin\theta + 2\sin\theta\cos\theta)$$

$$= 4\sin\theta(2\cos\theta - 1)$$

So

$$4\sin\theta(2\cos\theta - 1) = 0$$

Hence

$$\sin\theta = 0 \quad \text{or} \quad 2\cos\theta - 1 = 0$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, this gives

$$\theta = 0 \quad \text{or} \quad \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

At $\theta = 0$,

$$r = 4(1 - \cos 0) = 4(1 - 1) = 0$$

so this is the pole, but A is not the pole.

Therefore

$$\theta = \frac{\pi}{3}$$

Now find r:

$$r = 4\left(1 - \cos\frac{\pi}{3}\right) = 4\left(1 - \frac{1}{2}\right) = 2$$

To confirm the tangent is vertical, differentiate y:

$$y = 4(1 - \cos\theta)\sin\theta$$

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 4\big(\sin^2\theta + (1 - \cos\theta)\cos\theta\big)$$

At $\theta = \frac{\pi}{3}$,

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 4\left(\frac{3}{4} + \frac{1}{2}\cdot\frac{1}{2}\right)$$

$$= 4\left(\frac{3}{4} + \frac{1}{4}\right) = 4 \neq 0$$

So the tangent is indeed perpendicular to the initial line there.

Therefore the polar coordinates of A are

$$\left(2, \frac{\pi}{3}\right)$$

(b) Since $A = \left(2, \frac{\pi}{3}\right)$, its Cartesian x-coordinate is

$$x_A = r\cos\theta = 2\cos\frac{\pi}{3} = 1$$

So the tangent at A is the vertical line $x = 1$.

Hence the shaded region R is the area under the curve from O to A, above the initial line, so

$$A_R = \int y \, \mathrm{d}x = \int_0^{\pi/3} y \frac{\mathrm{d}x}{\mathrm{d}\theta} \, \mathrm{d}\theta$$

Using

$$y = 4(1 - \cos\theta)\sin\theta, \qquad \frac{\mathrm{d}x}{\mathrm{d}\theta} = 4\sin\theta(2\cos\theta - 1)$$

gives

$$A_R = 16 \int_0^{\pi/3} (1 - \cos\theta)\sin^2\theta(2\cos\theta - 1) \, \mathrm{d}\theta$$

Now use $\sin^2\theta = 1 - \cos^2\theta$:

$$\begin{aligned}(1 - \cos\theta)\sin^2\theta(2\cos\theta - 1) &= (1 - \cos\theta)(1 - \cos^2\theta)(2\cos\theta - 1) \\ &= (1 - \cos\theta - \cos^2\theta + \cos^3\theta)(2\cos\theta - 1) \\ &= 2\cos^4\theta - 3\cos^3\theta - \cos^2\theta + 3\cos\theta - 1\end{aligned}$$

Therefore

$$A_R = 16 \int_0^{\pi/3} \left(2\cos^4\theta - 3\cos^3\theta - \cos^2\theta + 3\cos\theta - 1\right) \mathrm{d}\theta$$

Use

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

so

$$\begin{aligned}\cos^4\theta &= \left(\frac{1 + \cos 2\theta}{2}\right)^2 \\ &= \frac{1}{4}\left(1 + 2\cos 2\theta + \cos^2 2\theta\right) \\ &= \frac{1}{4}\left(1 + 2\cos 2\theta + \frac{1 + \cos 4\theta}{2}\right) \\ &= \frac{1}{8}(3 + 4\cos 2\theta + \cos 4\theta)\end{aligned}$$

Also,

$$\int \cos^3\theta \, \mathrm{d}\theta = \int \cos\theta(1 - \sin^2\theta) \, \mathrm{d}\theta = \sin\theta - \frac{1}{3}\sin^3\theta$$

So

$$A_R = 16\left[\frac{3}{4}\theta + \frac{1}{2}\sin 2\theta + \frac{1}{16}\sin 4\theta - 3\left(\sin\theta - \frac{1}{3}\sin^3\theta\right)\right.$$
$$\left. - \left(\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right) + 3\sin\theta - \theta\right]_0^{\pi/3}$$
$$= 16\left[\frac{3}{4}\theta + \frac{1}{2}\sin 2\theta + \frac{1}{16}\sin 4\theta - 3\sin\theta + \sin^3\theta - \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta + 3\sin\theta - \theta\right]_0^{\pi/3}$$
$$= 16\left[\sin^3\theta + \frac{1}{4}\sin 2\theta + \frac{1}{16}\sin 4\theta - \frac{3}{4}\theta\right]_0^{\pi/3}$$

Now substitute the limits:

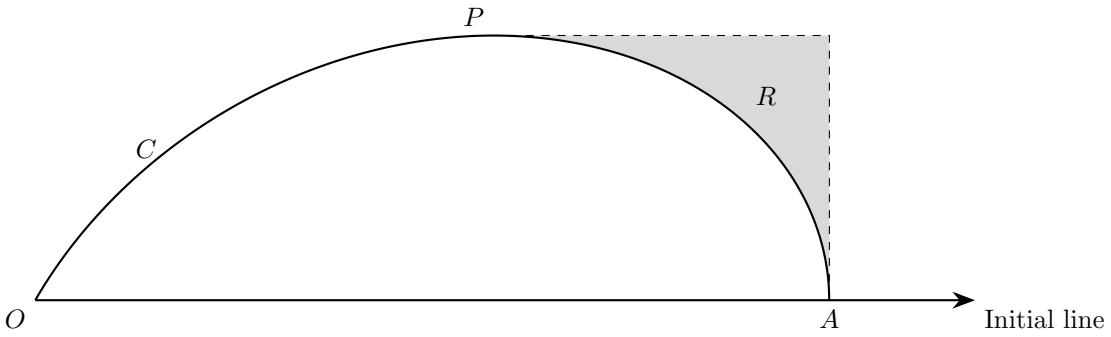
$$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin\frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$$

Hence

$$A_R = 16\left[\left(\frac{\sqrt{3}}{2}\right)^3 + \frac{1}{4}\cdot\frac{\sqrt{3}}{2} + \frac{1}{16}\cdot\left(-\frac{\sqrt{3}}{2}\right) - \frac{3}{4}\cdot\frac{\pi}{3}\right]$$
$$= 16\left[\frac{3\sqrt{3}}{8} + \frac{\sqrt{3}}{8} - \frac{\sqrt{3}}{32} - \frac{\pi}{4}\right]$$
$$= 16\left[\frac{15\sqrt{3}}{32} - \frac{\pi}{4}\right]$$
$$= \frac{15\sqrt{3}}{2} - 4\pi$$
$$= \frac{1}{2}(15\sqrt{3} - 8\pi)$$

Therefore the exact area of R is

$$\frac{1}{2}(15\sqrt{3} - 8\pi)$$



6. The curve C shown in the figure above has polar equation

$$r = 1 + 2\cos 2\theta, \qquad 0 \leq \theta \leq \frac{\pi}{3}$$

At the point P on C, the tangent to C is parallel to the initial line.

(a) Show that P has polar coordinates $\left(2, \frac{\pi}{6}\right)$. **[4]**

The point A on C has polar coordinates $(3, 0)$.

The finite region R, shown shaded in the figure above, is bounded by the arc AP of C, the tangent to C at P and the line through A parallel to $\theta = \frac{\pi}{2}$.

(b) Find the exact area of R. **[6]**

Solution

(a) Using

$$x = r\cos\theta, \qquad y = r\sin\theta$$

with

$$r = 1 + 2\cos 2\theta$$

we have

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}\theta}}{\frac{\mathrm{d}x}{\mathrm{d}\theta}}$$

Since the tangent is parallel to the initial line, it is horizontal, so

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0 \qquad \text{and} \qquad \frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$$

First,

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -4\sin 2\theta$$

Now

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}\theta} &= \frac{\mathrm{d}}{\mathrm{d}\theta}(r\sin\theta)\\ &= \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta\\ &= -4\sin 2\theta\sin\theta + (1+2\cos 2\theta)\cos\theta\\ &= -8\sin^2\theta\cos\theta + \big(1+2(1-2\sin^2\theta)\big)\cos\theta\\ &= -8\sin^2\theta\cos\theta + (3-4\sin^2\theta)\cos\theta\\ &= 3\cos\theta(1-4\sin^2\theta)\end{aligned}$$

So

$$3\cos\theta(1-4\sin^2\theta) = 0$$

For $0 \le \theta \le \frac{\pi}{3}$, we have $\cos\theta > 0$, so

$$1 - 4\sin^2\theta = 0$$

Hence

$$\sin\theta = \frac{1}{2}$$

and in the given interval this gives

$$\theta = \frac{\pi}{6}$$

Now find r:

$$r = 1 + 2\cos\Big(2\cdot\frac{\pi}{6}\Big) = 1 + 2\cos\frac{\pi}{3} = 1 + 2\cdot\frac{1}{2} = 2$$

Also,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(r\cos\theta) = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta$$

and at $\theta = \frac{\pi}{6}$,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = (-2\sqrt{3})\cdot\frac{\sqrt{3}}{2} - 2\cdot\frac{1}{2} = -3 - 1 = -4 \neq 0$$

so this does give a horizontal tangent.

Therefore P has polar coordinates

$$\Big(2, \frac{\pi}{6}\Big)$$

(b) Let Q be the point where the tangent at P meets the vertical line through A.

From part (a),

$$P = \Big(2, \frac{\pi}{6}\Big)$$

so in Cartesian coordinates,

$$P = \Big(2\cos\frac{\pi}{6},\, 2\sin\frac{\pi}{6}\Big) = (\sqrt{3}, 1)$$

Also $A = (3, 0)$, since its polar coordinates are $(3, 0)$.

The tangent at P is parallel to the initial line, so it is

$$y = 1$$

The line through A parallel to $\theta = \frac{\pi}{2}$ is vertical, so it is

$$x = 3$$

Hence

$$Q = (3, 1)$$

If the arc AP were replaced by the chord AP, the region would be triangle AQP. So

$$\text{Area}(R) = \text{Area}(\triangle AQP) - \text{area between chord } AP \text{ and arc } AP$$

First,

$$AQ = 1, \qquad QP = 3 - \sqrt{3}$$

so

$$\text{Area}(\triangle AQP) = \frac{1}{2}(1)(3 - \sqrt{3}) = \frac{3 - \sqrt{3}}{2}$$

Now find the area between the chord and the arc.

This is

$$\text{Area of sector } OAP - \text{Area}(\triangle OAP)$$

For the sector,

$$\text{Area of sector } OAP = \frac{1}{2}\int_0^{\pi/6} r^2 \, \mathrm{d}\theta = \frac{1}{2}\int_0^{\pi/6} (1 + 2\cos 2\theta)^2 \, \mathrm{d}\theta$$

Expand the square:

$$\begin{aligned}(1 + 2\cos 2\theta)^2 &= 1 + 4\cos 2\theta + 4\cos^2 2\theta \\ &= 1 + 4\cos 2\theta + 2(1 + \cos 4\theta) \\ &= 3 + 4\cos 2\theta + 2\cos 4\theta\end{aligned}$$

So

$$\begin{aligned}\text{Area of sector } OAP &= \frac{1}{2}\int_0^{\pi/6}(3+4\cos 2\theta+2\cos 4\theta)\,\mathrm{d}\theta\\ &= \frac{1}{2}\left[3\theta+2\sin 2\theta+\frac{1}{2}\sin 4\theta\right]_0^{\pi/6}\\ &= \frac{1}{2}\left(\frac{\pi}{2}+2\sin\frac{\pi}{3}+\frac{1}{2}\sin\frac{2\pi}{3}\right)\\ &= \frac{1}{2}\left(\frac{\pi}{2}+2\cdot\frac{\sqrt{3}}{2}+\frac{1}{2}\cdot\frac{\sqrt{3}}{2}\right)\\ &= \frac{1}{2}\left(\frac{\pi}{2}+\sqrt{3}+\frac{\sqrt{3}}{4}\right)\\ &= \frac{\pi}{4}+\frac{5\sqrt{3}}{8}\end{aligned}$$

For triangle OAP,

$$OA=3, \qquad OP=2, \qquad \angle AOP=\frac{\pi}{6}$$

so

$$\text{Area}(\triangle OAP)=\frac{1}{2}\cdot 3\cdot 2\cdot\sin\frac{\pi}{6}=\frac{3}{2}$$

Hence the area between chord AP and arc AP is

$$\left(\frac{\pi}{4}+\frac{5\sqrt{3}}{8}\right)-\frac{3}{2}$$

Therefore

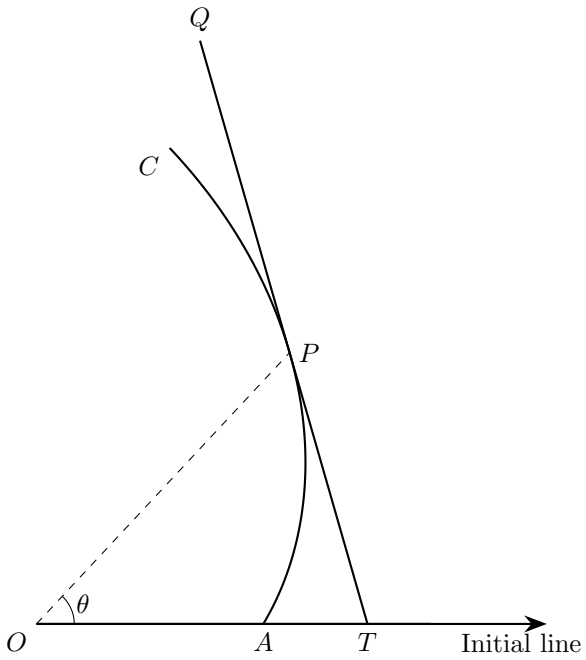
$$\begin{aligned}\text{Area}(R) &= \frac{3-\sqrt{3}}{2}-\left[\left(\frac{\pi}{4}+\frac{5\sqrt{3}}{8}\right)-\frac{3}{2}\right]\\ &= \frac{3-\sqrt{3}}{2}+\frac{3}{2}-\frac{\pi}{4}-\frac{5\sqrt{3}}{8}\\ &= 3-\frac{\pi}{4}-\frac{9\sqrt{3}}{8}\end{aligned}$$

So the exact area of R is

$$3-\frac{\pi}{4}-\frac{9\sqrt{3}}{8}$$

equivalently

$$\frac{24-2\pi-9\sqrt{3}}{8}$$



7. The diagram shows part of a spiral C.

The point P has polar coordinates (r, θ), where $0 \leq \theta \leq \frac{\pi}{2}$.

The tangent at P is the line TPQ. The point A lies on C and on the initial line, with $OA = 2$.

Let ψ be the angle that TPQ makes with the initial line.

(a) By expressing the Cartesian coordinates of P in terms of r and θ, show that

$$\tan\psi = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} \qquad \textbf{[4]}$$

(b) The spiral is equiangular: at every point P, the acute angle between OP and TPQ is a constant α, where $0 < \alpha < \frac{\pi}{2}$.

Use the result of part (a) to show that the polar equation of C is

$$r = 2e^{(\cot\alpha)\theta} \qquad \textbf{[7]}$$

Solution

(a) Let the Cartesian coordinates of P be (x, y). Since P has polar coordinates (r, θ),

$$x = r\cos\theta, \qquad y = r\sin\theta$$

Differentiate both with respect to θ:

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta$$

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta$$

The angle ψ is the angle the tangent makes with the initial line, so

$$\tan\psi = \frac{\mathrm{d}y}{\mathrm{d}x}$$

Using

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}\theta}}{\frac{\mathrm{d}x}{\mathrm{d}\theta}}$$

gives

$$\tan\psi = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta}$$

as required.

(b) Let

$$r' = \frac{\mathrm{d}r}{\mathrm{d}\theta}$$

From part (a),

$$\tan\psi = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$$

Divide top and bottom by $r'\cos\theta$:

$$\tan\psi = \frac{\tan\theta + \frac{r}{r'}}{1 - \tan\theta\frac{r}{r'}}$$

Now let β be defined by

$$\tan\beta = \frac{r}{r'}$$

Then

$$\begin{aligned}\tan\psi &= \frac{\tan\theta + \tan\beta}{1 - \tan\theta\tan\beta} \\ &= \tan(\theta + \beta)\end{aligned}$$

So the angle between OP and the tangent is β.

But the spiral is equiangular, so this acute angle is the constant α. Hence

$$\beta = \alpha$$

and therefore

$$\tan\alpha = \frac{r}{r'}$$

So

$$\frac{r}{\frac{\mathrm{d}r}{\mathrm{d}\theta}} = \tan\alpha$$

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = r\cot\alpha$$

This is a separable differential equation:

$$\frac{1}{r}\,\mathrm{d}r = \cot\alpha\,\mathrm{d}\theta$$

Integrating,

$$\int \frac{1}{r}\,\mathrm{d}r = \int \cot\alpha\,\mathrm{d}\theta$$

$$\ln r = (\cot\alpha)\theta + c$$

At the point A, the curve is on the initial line and $OA = 2$, so

$$\theta = 0, \qquad r = 2$$

Substitute into the equation:

$$\ln 2 = (\cot\alpha)(0) + c$$

$$c = \ln 2$$

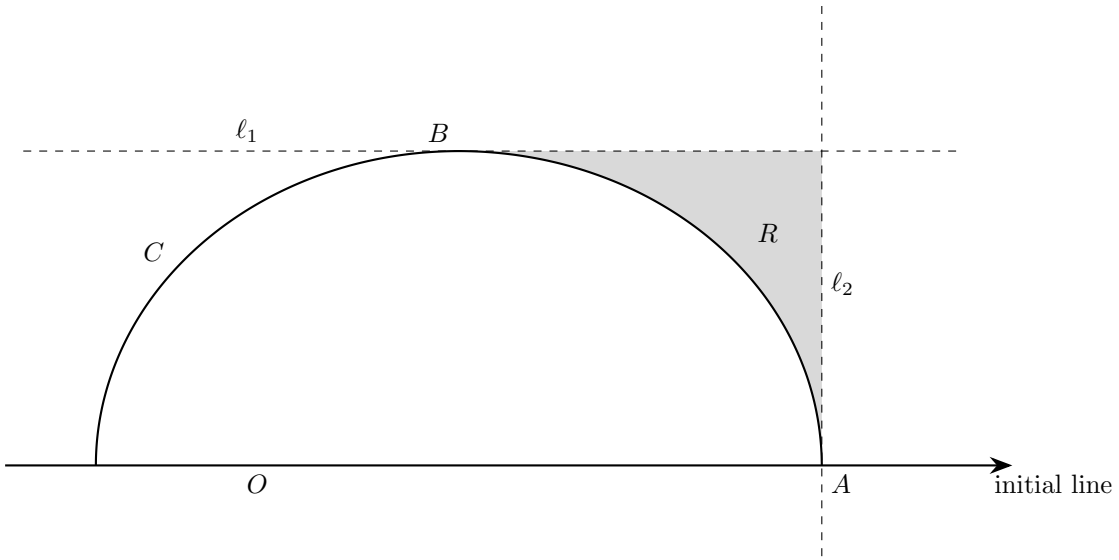
Hence

$$\ln r = (\cot\alpha)\theta + \ln 2$$

$$r = 2e^{(\cot\alpha)\theta}$$

Therefore the polar equation of C is

$$r = 2e^{(\cot\alpha)\theta}$$



8. The figure above shows a sketch of the curve C with equation

$$r = \frac{6}{2 - \cos\theta} \qquad 0 \leq \theta \leq \pi$$

Given that C meets the initial line at the point A, as shown in the figure above

(a) write down the polar coordinates of A. **[1]**

The line ℓ_1, also shown in the figure above, is the tangent to C at the point B and is parallel to the initial line.

(b) Use calculus to determine the polar coordinates of B. **[6]**

The line ℓ_2, also shown in the figure above, is the tangent to C at A and is perpendicular to the initial line.

The region R, shown shaded in the figure above, is bounded by C, ℓ_1 and ℓ_2.

(c) By first obtaining a Cartesian equation for C, use algebraic integration to find the exact area of R, giving your answer in the form $\sqrt{3}(p + q\pi)$ where p and q are constants to be determined. **[10]**

Solution

(a) On the initial line, $\theta = 0$.

$$r = \frac{6}{2 - \cos 0} = \frac{6}{2 - 1} = 6$$

So the polar coordinates of A are $(6, 0)$.

(b) Using

$$x = r\cos\theta, \qquad y = r\sin\theta$$

we have

$$x = \frac{6\cos\theta}{2-\cos\theta}, \qquad y = \frac{6\sin\theta}{2-\cos\theta}$$

For a tangent parallel to the initial line, we need

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 0$$

and

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\dfrac{\mathrm{d}y}{\mathrm{d}\theta}}{\dfrac{\mathrm{d}x}{\mathrm{d}\theta}}$$

Now differentiate x and y with respect to θ.

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}\theta} &= \frac{(-6\sin\theta)(2-\cos\theta) - 6\cos\theta(\sin\theta)}{(2-\cos\theta)^2} \\ &= \frac{-12\sin\theta + 6\sin\theta\cos\theta - 6\sin\theta\cos\theta}{(2-\cos\theta)^2} \\ &= \frac{-12\sin\theta}{(2-\cos\theta)^2}\end{aligned}$$

For $0 < \theta < \pi$, $\sin\theta \neq 0$, so $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$. Hence a horizontal tangent occurs when

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$$

Now

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}\theta} &= \frac{6\cos\theta(2-\cos\theta) - 6\sin^2\theta}{(2-\cos\theta)^2} \\ &= \frac{12\cos\theta - 6\cos^2\theta - 6\sin^2\theta}{(2-\cos\theta)^2} \\ &= \frac{12\cos\theta - 6(\cos^2\theta + \sin^2\theta)}{(2-\cos\theta)^2} \\ &= \frac{12\cos\theta - 6}{(2-\cos\theta)^2} \\ &= \frac{6(2\cos\theta - 1)}{(2-\cos\theta)^2}\end{aligned}$$

So

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0 \quad \Rightarrow \quad 2\cos\theta - 1 = 0 \quad \Rightarrow \quad \cos\theta = \frac{1}{2}$$

Since $0 \leq \theta \leq \pi$,

$$\theta = \frac{\pi}{3}$$

Then

$$r = \frac{6}{2-\cos\left(\frac{\pi}{3}\right)} = \frac{6}{2-\frac{1}{2}} = \frac{6}{\frac{3}{2}} = 4$$

So the polar coordinates of B are

$$(4, \tfrac{\pi}{3})$$

(c) Start with the polar equation

$$r = \frac{6}{2 - \cos\theta}$$

Rearranging,

$$r(2 - \cos\theta) = 6$$

Using $x = r\cos\theta$, this gives

$$2r - x = 6$$

so

$$2r = x + 6$$

Squaring both sides and using $r^2 = x^2 + y^2$,

$$4r^2 = (x+6)^2$$

$$4(x^2 + y^2) = x^2 + 12x + 36$$

$$3x^2 + 4y^2 - 12x - 36 = 0$$

Complete the square in x:

$$3(x^2 - 4x) + 4y^2 = 36$$

$$3\big((x-2)^2 - 4\big) + 4y^2 = 36$$

$$3(x-2)^2 + 4y^2 = 48$$

So a Cartesian equation for C is

$$3(x-2)^2 + 4y^2 = 48$$

Since $0 \le \theta \le \pi$, the curve lies above the x-axis, so

$$y = \frac{\sqrt{3}}{2}\sqrt{16 - (x-2)^2}$$

Now find the bounding lines.

At $A = (6, 0)$, the tangent ℓ_2 is perpendicular to the initial line, so it is vertical:

$$\ell_2 : \ x = 6$$

From part (b), $B = (4, \pi/3)$, so in Cartesian coordinates

$$x = 4\cos\frac{\pi}{3} = 2, \qquad y = 4\sin\frac{\pi}{3} = 2\sqrt{3}$$

Since ℓ_1 is parallel to the initial line, it is horizontal:

$$\ell_1 : \ y = 2\sqrt{3}$$

Hence the area of R is

$$\int_2^6 \left(2\sqrt{3} - \frac{\sqrt{3}}{2}\sqrt{16-(x-2)^2}\right) \mathrm{d}x$$

So

$$\begin{aligned}\text{Area}(R) &= \int_2^6 2\sqrt{3}\,\mathrm{d}x - \frac{\sqrt{3}}{2}\int_2^6 \sqrt{16-(x-2)^2}\,\mathrm{d}x \\ &= 8\sqrt{3} - \frac{\sqrt{3}}{2}\int_2^6 \sqrt{16-(x-2)^2}\,\mathrm{d}x\end{aligned}$$

Let

$$u = x - 2$$

Then when $x = 2$, $u = 0$, and when $x = 6$, $u = 4$. So

$$\text{Area}(R) = 8\sqrt{3} - \frac{\sqrt{3}}{2}\int_0^4 \sqrt{16-u^2}\,\mathrm{d}u$$

Now use $u = 4\sin t$. Then

$$\mathrm{d}u = 4\cos t\,\mathrm{d}t$$

and the limits become $t = 0$ to $t = \frac{\pi}{2}$.

$$\begin{aligned}\int_0^4 \sqrt{16-u^2}\,\mathrm{d}u &= \int_0^{\pi/2} \sqrt{16-16\sin^2 t}\,(4\cos t)\,\mathrm{d}t \\ &= \int_0^{\pi/2} 4\cos t \cdot 4\cos t\,\mathrm{d}t \\ &= 16\int_0^{\pi/2} \cos^2 t\,\mathrm{d}t \\ &= 16\int_0^{\pi/2} \frac{1+\cos 2t}{2}\,\mathrm{d}t \\ &= 8\int_0^{\pi/2} (1+\cos 2t)\,\mathrm{d}t \\ &= 8\left[t + \frac{1}{2}\sin 2t\right]_0^{\pi/2} \\ &= 8\left(\frac{\pi}{2}\right) \\ &= 4\pi\end{aligned}$$

Therefore

$$\begin{aligned}\text{Area}(R) &= 8\sqrt{3} - \frac{\sqrt{3}}{2}(4\pi) \\ &= 8\sqrt{3} - 2\pi\sqrt{3} \\ &= \sqrt{3}(8-2\pi)\end{aligned}$$

So the exact area is

$$\sqrt{3}(8-2\pi)$$

and therefore

$$p = 8, \qquad q = -2$$